Computer Vision:

Player Tracking in Sports - Multi-View Tracking and 3D Reconstruction

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Contents

Introduction

- **Goal:** track players and the ball, then reconstruct their 3D trajectories.
- **Challenges:** ball detection, fast motion, mutual occlusions.
- **Setup:** three synchronised, calibrated cameras.

Detection: Prior Work and Limitations

- **MOG2 background subtraction** [3] (with CLAHE [9]):
  - Floor specularities and shadows survived the shadow detection.
  - **False positives:** moving spectators produce false positives.
  - **False negatives:** standing referees and players may be absorbed into the background model.
- **Pre-trained YOLO26** [5]:
  - Correctly computes bounding boxes for players and the ball.
  - Output is a simple labelled bounding box, no identity information.
  - Need a post-processing step to assign identities.

Detection: Fine - Tuned YOLO26

- Fine-tuned on ∼500 hand-labelled frames from the same game (varied actions and camera views).
- Per-frame identity comes straight from detection.
- Ball detection is still inconsistent.

Detection: Improvements

- **Dual-pass inference** (the ball occupies few pixels):
  - Primary pass tuned for players.
  - High-resolution and low-confidence pass for ball recall.
  - Both outputs merged into one set of detections.
- **Class-agnostic NMS** [4] (one player, several overlapping boxes):
  - Rank by confidence; drop boxes overlapping a kept one (IoU).
  - Keeps the most confident box per player for the tracker.

Tracking: Prior Work and Limitations

- **SORT** [1] (tracking by detection):
    - Kalman filter with a constant-velocity motion model.
    - Hungarian assignment on IoU overlap to match detections to tracks.
    - No appearance cue: frequent identity switches when players cross paths.

Final Tracking: DeepSORT

DeepSORT [6] adds appearance tracking to solve ID switching.

- **Visual Features:** Integrates an OSNet [8] appearance embedding.
- **Two-Stage Matching:** Prioritizes appearance similarity, falling back to overlap.
- **Appearance Gallery:** Maintains a history of embeddings for each track to compare against new detections.
- **Result:** Robust tracking through close interactions and partial occlusions.

Tracking: Custom Improvements

Specific enhancements made to the baseline DeepSORT pipeline:

- **NSA Kalman Update:** Scales the measurement noise by detection confidence to drastically reduce bounding box jitter.
- **Low-Confidence Recovery:** Uses ByteTrack [7] logic to track heavily occluded players via weak detections.
- **Track Resurrection:** Matches incoming detections against lost-track galleries to re-identify players re-entering the frame.
- **Coasting:** Emit tracks prediction when no detection is available, to avoid losing players during occlusions.
- **Label Resolution:** Smooths player classifications over time using a confidence-weighted majority vote.

3D Reconstruction: Camera Calibration

- **World frame:** a standard FIBA court model (true court dimensions in mm; origin at court centre).
- **Intrinsics:** known a priori for each camera.
- **Extrinsics:** estimated by solving Perspective-n-Point (PnP).
- PnP maps manually annotated court landmarks to their 3D world coordinates.

3D Reconstruction: Triangulation

Players and the ball are treated differently:

- **Players:** a planar homography maps the foot pixel onto court-floor coordinates.
  - Floor constraint allows single-view localisation under occlusion.
  - Estimates fused across the three cameras after discarding outliers.
- **Ball:** airborne, so triangulated from at least 2 views.
  - DLT stacks the per-camera ray constraints; SVD solves the system.
  - High-error views discarded and the point re-triangulated.

3D Reconstruction: Trajectory Smoothing

Remove residual noise from detection jitter and calibration error.

- Forward Kalman filter, then a Rauch–Tung–Striebel backward pass.
- Past and future observations correct each state.
- Reject implausible spatial jumps as outliers.
- Split a trajectory at long gaps, rather than interpolate across frames with no detections.

Evaluation Metrics

- **Detection:** precision, recall, F_1 (true positive at IoU > 0.5).
- **Tracking:** IDF1 for identity; **HOTA** [2] as the primary metric.
- **3D geometry:** reprojection error (mm), reflecting calibration and triangulation.
- **Trajectory:** ADE, FDE, MTE (mm) of reprojection against ground-truth 2D data.

Findings: Detection and Tracking

- DetA $\approx$ AssA: errors are missed players, not bad matches.
- Detection drives tracking quality.
- The ball is the main bottleneck from qualitative review.

Cam	Prec.	Rec.	F_1	IoU
cam_13	0.872	0.732	0.796	0.758
cam_2	0.835	0.761	0.796	0.757
cam_4	0.923	0.822	0.870	0.796

Cam	IDF1	HOTA	DetA	AssA
cam_13	0.569	0.434	0.469	0.405
cam_2	0.610	0.480	0.453	0.511
cam_4	0.638	0.521	0.553	0.495

Findings: 3D and Trajectories

- Metre-scale trajectory errors from calibration bias.
- Largest for cam_13: wide overview, players farthest, more mm per pixel.
- Similar ADE/FDE/MTE: a constant offset, not growing drift.
- Low fragments; RMSE $\gg$ median: a few outlier frames dominate.

Cam	ADE	FDE	MTE	Frag.
cam_13	9976.1	8503.9	10662.2	2
cam_2	3984.4	5098.8	4373.3	5
cam_4	8636.7	7238.4	8739.9	5

Cam	Mean	Median	RMSE
cam_13	499.6	300.5	844.9
cam_2	772.5	459.6	1378.8
cam_4	395.3	176.0	905.4

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Appendix: Dataset and Fine-Tuning

- **Clip:** Sanbapolis basketball, three synchronised cameras (cam_13, cam_2, cam_4).
- **Length:** 525 frames at 25 fps (≈ 21 s).
- **Ground truth:** 2D boxes annotated every fifth frame; intrinsics provided.
- **Detector training:** separate disjoint clip, from YOLO26m weights.
- **Schedule:** up to 300 epochs (patience 30), 1280 px input, batch size 4.

Appendix: Metric Formulas (Detection and Tracking)

Detection

$$\text{Prec} = \frac{TP}{TP+FP}, \quad \text{Rec} = \frac{TP}{TP+FN}, \quad F_1 = \frac{2\,\text{Prec}\cdot\text{Rec}}{\text{Prec}+\text{Rec}}, \quad \text{IoU} = \frac{|A\cap B|}{|A\cup B|}$$

Tracking (TP, FP, FN from matching at localization threshold α; HOTA integrates over α)

$$\text{IDF1} = \frac{2\,IDTP}{2\,IDTP + IDFP + IDFN}, \qquad \text{HOTA} = \int_0^1 \sqrt{\text{DetA}_\alpha \cdot \text{AssA}_\alpha}\; d\alpha$$

$$\text{DetA}_\alpha = \frac{|TP|}{|TP|+|FN|+|FP|}, \qquad \text{AssA}_\alpha = \frac{1}{|TP|}\sum_{c\in TP} A(c)$$

$$A(c) = \frac{|TPA(c)|}{|TPA(c)|+|FNA(c)|+|FPA(c)|}, \qquad \text{LocA} = \frac{1}{|TP|}\sum_{c\in TP} \text{IoU}(c)$$

Appendix: Metric Formulas (3D and Trajectory)

Reprojection error (3D point X_i projected by camera P, observed at x_i)

$$e_i = \|x_i - \pi(P, X_i)\|, \qquad \text{RMSE} = \sqrt{\tfrac{1}{N}\textstyle\sum_i e_i^2}$$

Trajectory error (prediction p_t, ground truth g_t, over T frames)

$$\text{ADE} = \frac{1}{T}\sum_{t=1}^{T}\|p_t - g_t\|, \quad \text{FDE} = \|p_T - g_T\|, \quad \text{MTE} = \mathrm{median}_t\|p_t - g_t\|$$

Appendix: Pipeline Parameters (Detection)

MOG2 and preprocessing

Parameter	Value
Learning rate	-1 (adaptive)
Detect shadows	off
CLAHE clip limit	6.0
CLAHE tile grid	8×8
Min. blob area (px^2)	500
Max. blob area (px^2)	10000

YOLO detection and NMS

Parameter	Value
Single-pass	
Confidence	0.3
Inference size	640
Built-in NMS IoU	0.45
Two-pass fine-tuned	
Player size (px)	1280
Ball size (px)	1600
Ball confidence	0.2
Merge NMS IoU	0.75

Appendix: Pipeline Parameters (Tracking and Geometry)

Tracking (DeepSORT)

Parameter	Value
Max. IoU distance	0.7
Max. appearance distance	0.2
Max. age (frames)	60
Confirmation hits	2
Feature budget	100
Coast age (frames)	5
Resurrection window	150
OSNet embedding dim	512

Geometry

Parameter	Value
Max. interpolated gap	5
EMA weight α	0.5
Ground inlier gate (mm)	600
Ball reproj. gate (px)	30
Minimum views	2
Mahalanobis gate (χ^2)	11.345
Position noise (mm)	30
Measurement noise (mm)	80

Appendix: Detection Results

Method	Camera	Precision	Recall	F_1	Mean IoU
MOG2	cam_13	0.471	0.262	0.337	0.651
	cam_2	0.345	0.408	0.374	0.663
	cam_4	0.328	0.293	0.309	0.700
YOLO pre-trained	cam_13	0.851	0.617	0.716	0.754
	cam_2	0.858	0.778	0.816	0.782
	cam_4	0.922	0.801	0.857	0.808
YOLO fine-tuned	cam_13	0.872	0.732	0.796	0.758
	cam_2	0.835	0.761	0.796	0.757
	cam_4	0.923	0.822	0.870	0.796

Appendix: Tracking Results

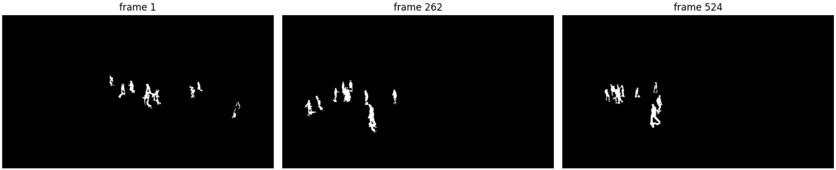
Tracker	Camera	IDP	IDR	IDF1	HOTA	DetA	AssA	LocA
SORT	cam_13	0.660	0.514	0.590	0.438	0.484	0.398	0.797
	cam_2	0.743	0.607	0.668	0.517	0.515	0.522	0.794
	cam_4	0.768	0.605	0.679	0.542	0.575	0.514	0.829
DeepSORT	cam_13	0.588	0.533	0.569	0.434	0.469	0.405	0.769
	cam_2	0.617	0.603	0.610	0.480	0.453	0.511	0.761
	cam_4	0.670	0.605	0.638	0.521	0.553	0.495	0.798

Appendix: 3D Geometry Errors

	Reprojection error						Trajectory error		
Camera	Mean	Median	RMSE	Acc@25	Acc@50	Acc@100	ADE	FDE	MTE
cam_13	499.6	300.5	844.9	0.087	0.134	0.219	9976.1	8503.9	10662.2
cam_2	772.5	459.6	1378.8	0.023	0.038	0.076	3984.4	5098.8	4373.3
cam_4	395.3	176.0	905.4	0.188	0.247	0.359	8636.7	7238.4	8739.9

Errors in mm; Acc columns are fractions at 25, 50, 100 mm. Fragment counts: 2, 5, 5.

Appendix: Qualitative Outputs, Detection (1/2)



MOG2 foreground mask

Pre-trained YOLO26 detections

Appendix: Qualitative Outputs, Detection (2/2)

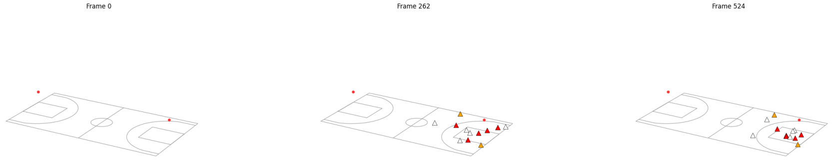
Fine-tuned YOLO26, merged dual-pass detections

Appendix: Qualitative Outputs, Tracking

SORT tracks

DeepSORT tracks after label resolution

Appendix: Qualitative Outputs, 3D Reconstruction



Smoothed 3D reconstruction